

allowing several users to connect to a single light source and access data. Each Li-Fi light consumes just 10 watts of power, which is comparable to a small LED bulb. To provide connectivity, you just replace the standard lights with Li-Fi-enabled ones.

While Wi-Fi can cover multiple floors, its speed often on the other floors. To ensure reliable connectivity, multiple Li-Fi light sources must be installed across an office, similar to how Wi-Fi routers are spaced out in enterprise settings to ensure complete coverage.

How does Li-Fi perform across multiple floors?

Unlike Wi-Fi, Li-Fi requires a denser deployment to provide full coverage across different floors. Every light source in the building, regardless of the floor, acts as an access point, offering internet connectivity to users. This approach enhances the overall user experience and improves the average speed per user compared to Wi-Fi.

How does Li-Fi transmit data without altering regular light appearance?

To the user, the light appears just like any other daily light. However, we modulate a signal that carries data within that light. This signal is combined with the light and transmitted, and on our receiver, we decode it. Essentially, we mix the signal with the light, which carries the data, and decode it on the receiving end. From a user's perspective, it looks and functions like an ordinary light; you cannot tell the difference. For the user, Li-Fi light looks just like regular light. But inside that light, we embed a modulated signal carrying data. Our receiver decodes it seamlessly, so while you are using data through light, the visual experience remains unchanged.

Who are the primary target customers for outdoor and indoor Li-Fi solutions? Why should they consider Li-Fi?

Outdoor Li-Fi solutions, like our LC Link, are primarily targeted

at telecom service providers and internet service providers (ISPs). In India, companies like Airtel, Jio, BSNL, and Vodafone are potential customers, as well as smaller local internet service providers (ISPs) nationwide. These ISPs often face challenges in laying fibre cables up to users' homes, making them ideal customers. Additionally, the defence sector, particularly in border areas where connectivity is challenging, also benefits from this product.

As for indoor Li-Fi, since it is currently more expensive than Wi-Fi, it is not yet targeted at home users. Instead, we focus on enterprises that require more secure networks and higher speeds. We have deployed Li-Fi solutions in several customer offices, converting them into Li-Fi-enabled environments. These businesses use our Li-Fi-enabled lights and dongles, which connect to laptops and devices, allowing them to access the internet securely without relying on Wi-Fi. While telecom operators and ISPs are the main customers for LC Link, enterprises are the key customers for our indoor Li-Fi solutions.

We have deployed Li-Fi in several offices, converting them into fully Li-Fi-enabled environments. These businesses use our lights and dongles, which connect to laptops and devices, allowing secure internet access without Wi-Fi dependency. While telecom operators and ISPs are primary customers for LC Link, enterprises are the main users of our indoor Li-Fi.

Can you share specific use cases of Li-Fi technology with us?

Traditional options like Wi-Fi, Bluetooth, and RF mesh devices often face security vulnerabilities and interference issues in industrial environments, leading many industries to rely on cable infrastructure despite its maintenance challenges. Li-Fi addresses these issues by offering secure, reliable wireless connectivity

without cables. We have already implemented Li-Fi in some industrial setups and are expanding this with large manufacturing companies.

Another critical use case for Li-Fi is in defence and government offices, where Wi-Fi is often prohibited due to security concerns. Li-Fi's signal remains within a closed environment, making it a highly secure alternative to LAN cables. We are actively working with defence forces to integrate Li-Fi as a secure network within their offices.

Our LC Link product can be used for tower-to-tower communication, known as backhaul in 4G and 5G networks, improving telecom infrastructure. Additionally, LC Link can solve last-mile connectivity issues in rural and remote areas by replacing the need for fibre cables. Instead of laying cables to each home, operators can install a device on a rooftop to distribute the internet to multiple customers, making broadband access more feasible and cost-effective. This approach is valuable for extending connectivity in underserved regions.

How can Li-Fi function effectively in jammed areas?

Li-Fi works despite typical jammers. Jammers are typically employed to prevent outside interference, and Li-Fi offers a localised, secure network solution. It can be used as an alternative in events, conferences, and other situations where a secure network is needed.

What happens if there is a line-of-sight obstruction, such as someone standing between the laptop and the Li-Fi panel on the ceiling?

Li-Fi relies on a direct line of sight between the light source and the laptop. If the connection is completely obstructed, the signal will be disrupted, leading to a loss of connection. However, ongoing research is exploring ways to enable Li-Fi networks to function in non-line-of-sight situations by using light reflections from walls and other surfaces within the room. **EFY**